

Ashvath V Cheppalli

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EDUCATION

Northeastern University | *Boston, MA*

Sep 2025 - Present

Master of Science, Artificial Intelligence

GPA: 3.83

- **Coursework:** Foundation of AI, Algorithms, Applied Programming & Data Processing for AI, Machine Learning, Neural Networks, Natural Language Processing, Pattern Recognition and Computer Vision

Rajalakshmi Engineering College | *Chennai, India*

2021 - 2025

Bachelor of Technology, AI and Machine Learning

GPA: 7.49/10

- **Coursework:** Deep Learning, Computer Vision, Reinforcement Learning, Databases, Neural Networks, Data Structures & Algorithms, Statistics & Probability, Object Detection, Operating Systems, Linear Algebra, Discrete Mathematics

TECHNICAL SKILLS

- **Languages:** Python, HTML, CSS, JavaScript, SQL
- **ML/AI:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, OpenCV, YOLOv8, LangChain
- **Data:** Pandas, NumPy, Matplotlib, Seaborn, Plotly
- **Databases:** MySQL, PostgreSQL, MongoDB, Firebase
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Google Cloud, Docker, Git, GitHub, CI/CD
- **Tools:** VS Code, FastAPI, Flask, Node.js, React, REST APIs
- **Soft Skills:** Problem Solving, Technical Communication, Cross-functional Collaboration

PROJECT EXPERIENCE

Layer-Aware Quantization Pipeline for Efficient DistilBERT Deployment

[Project Link](#)

- Built quantization pipeline for **DistilBERT** achieving **2.19x** speedup and **64%** memory reduction using **INT8-PTQ**.
- Implemented layer-wise sensitivity analysis across **20 transformer** components on **SST-2** for precision deployment.
- Benchmarked **4 precision modes** on accuracy, latency and robustness with **24 automated pytest** unit tests.

Intelligent Human Detection for Emergency Fire Evacuation

[Project Link](#)

- Fine-tuned **YOLOv8** on **4,000+** thermal frames achieving **89.6% mAP50** and **86.7% precision** in fire scenarios.
- Launched **Flutter** mobile app with **YOLOv8** at **24 FPS**, enabling real-time post-incident rescue analysis.
- Curated and annotated **4,000+ thermal frames** across **12 fire scenarios**, improving model generalization by **18%**.

AI-Optimized Energy Consumption Forecasting System

[Project Link](#)

- Trained **XGBoost regression** model on **1M+** records, achieving **0.038 kWh MAE**, a **95%** improvement over baseline.
- Executed **EDA** and feature engineering in **Python** and **Pandas**, uncovering peak consumption patterns for optimization.
- Deployed forecasting model as a **REST API** endpoint, serving real-time predictions at under **120ms** latency.

WORK EXPERIENCE

Skillible | *Chennai, India*

Jul 2024 - Aug 2024

AI Intern

- Investigated **6+** bias-mitigation strategies for generative **AI** models including re-weighting and adversarial debiasing.
- Evaluated **transformer** model performance across **NLP** tasks using **Python** and **Scikit-learn** to guide model selection.
- Developed **15+** prompt templates using advanced **prompt engineering** techniques to optimize **LLM** output quality.

Anugraha Systems (Rajalakshmi Engineering College) | *Chennai, India*

Jan 2023

Web Development Intern

- Built end-to-end web applications using **React** and **MySQL** across the full software development lifecycle.
- Redesigned frontend-to-backend data flow for **4** internal workflows, improving overall system responsiveness.
- Shipped a customer-facing **dashboard** with real-time data display, improving **UI** responsiveness.